

Real-Time Virtual Human for Promoting Clinical Trial Education and Recruitment

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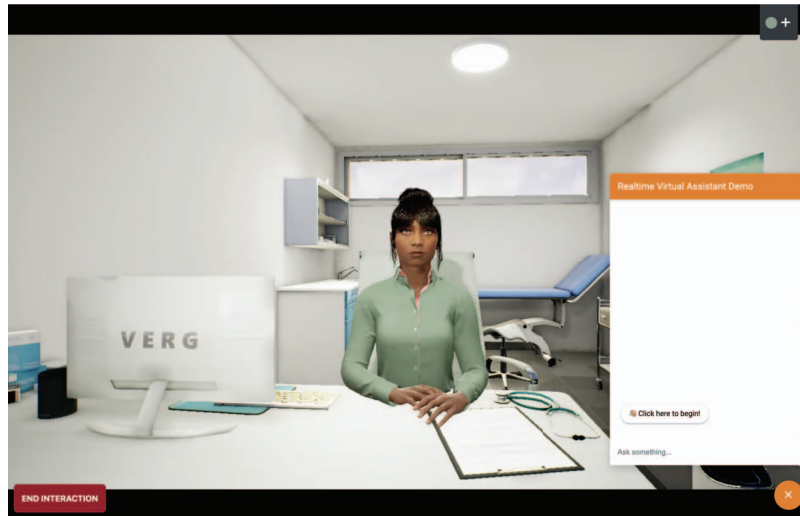


Figure 1: Real-time desktop VR virtual human running in web browser using Metahumans, Pixelstreaming, Dialogflow, and ChatGPT

ABSTRACT

Virtual human technology has gained popularity in the healthcare domain due to their cost-effectiveness, accessibility from computers and mobile devices, and ability to be tailored and customized. However, many virtual human applications rely on pre-scripted interactions and lack real-time capabilities. This limits the potential of virtual human technology in areas such as clinical trial recruitment, where there is a need to deliver up-to-date information on relevant clinical trials and tailor to diverse populations with diverse needs. In this work, we develop a virtual human desktop VR architecture that allows for real-time interactions. We demonstrate the real-time, tailored virtual human abilities such as dynamic responses to unscripted user input and assisting in searching for clinical trials based on user demographics.

Keywords: human-centered computing, virtual humans, real-time

Index Terms: Human-centered computing—Human computer interaction (HCI)—Real-time system architecture; Computer systems organization—Real-time systems—Interactive systems and tools

1 INTRODUCTION

Virtual human technology has emerged as a promising solution in the healthcare domain [10] [12] [11]. It offers cost-effective and accessible platforms such as desktop and mobile for tailored and customizable interventions [13]. Despite these advantages, a

significant limitation in many virtual human applications lies in their reliance on pre-scripted interactions, hindering real-time capabilities. This limitation becomes particularly evident in critical domains such as clinical trial recruitment, where the need for up-to-date information and personalized interactions with diverse populations is paramount.

Our work addresses the constraints of pre-scripted virtual human interactions that has the potential to support the needs of healthcare applications such as clinical trials recruitment. We use emerging technologies such as Unreal Engine [8] and Metahumans [5] for high-fidelity virtual humans and streaming capabilities, and Google Dialogflow [3] and Open AI's Chat GPT [6] for discourse management and natural language processing. In this paper, we present how our virtual human intervention architecture utilizes these technologies and showcase the real-time capabilities they achieve. We emphasize the system's ability to respond dynamically to unscripted user input, providing a more informative and engaging user experience. Furthermore, we demonstrate its utility in assisting users in the search for clinical trials by tailoring recommendations based on user demographics. Through this research, we aim to contribute to the advancement of virtual human technology, unlocking new possibilities for real-time, personalized interactions in healthcare applications.

2 SYSTEM ARCHITECTURE

Our virtual human architecture leverages a synergistic integration of advanced technologies to deliver a high-fidelity and real-time virtual human experience. The key technologies include Unreal Engine and Metahumans for rendering realistic virtual humans, Google Dialogflow for dialog management, and OpenAI's Chat GPT for handling dynamic user input. Figure 2 summarizes how these technologies work together to achieve our real-time virtual human intervention.

Rendering and Real-Time Streaming. The virtual human ap-

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plication is developed using Unreal Engine 4.27. The application uses Unreal Engine's Metahumans for high-fidelity virtual humans. Real-time capabilities are achieved by Unreal Engine's Pixel Streaming feature, which allows a packaged version of an application to be hosted on and streamed from a desktop PC or cloud server [9]. Users connect to the real-time stream via a standard web browser.

Speech and Lip Syncing. Within the Unreal Engine application, real-time speech and lip syncing are accomplished using the Google Text To Speech plugin [4] and the OVR lipsync plugin [7]. The virtual human generates speech based on text received through Pixel Streaming from the front-end files. These files, responsible for interfacing with Google Dialogflow, handle dynamic interactions between the user and virtual human.

Dialog Management and Dynamic Responses. Google Dialogflow acts as the primary dialog manager, employing a mostly scripted structure to guide user conversations. Dialogflow handles gathering information about the user via slot filling, a natural language processing technique to extract specific pieces of information during a conversation. This information is later used to query clinicaltrials.gov [1] based on user information in order to provide tailored information about relevant clinical trials. Google Cloud Functions [2] facilitates the API call made to clinicaltrials.gov. Additionally, before displaying clinical trial information to the user, it is run through Open AI's Chat GPT in order to summarize and present the information in a tailored, more accessible manner (for example, by lowering the health literacy).

In order to allow for dynamic conversation, API calls to Open AI's Chat GPT are also made from Google Cloud Functions. This enables the virtual human to handle user input outside the Dialogflow script. A predefined knowledge base containing data from clinical trial resources (e.g., NIH) is utilized to constrain agent responses to clinical trials. A vector similarity of the user's query and the knowledge base entries is performed to determine what information to use to answer the query. If a query is out of scope or unable to be answered, the system provides a default response.

User Interaction and Accessibility. The resulting desktop VR application is accessible via a web browser (see Figure 2). Users engage with the virtual human through a chat box, enabling them to input responses via text or select predefined buttons. This interactive approach ensures a seamless and immersive experience, enhancing user engagement and interaction with the virtual environment.

3 CONCLUSION

In this work, we have presented a virtual human architecture designed to address the limitations of pre-scripted interactions, particularly in the context of healthcare applications such as clinical trial recruitment. Leveraging a sophisticated blend of Unreal Engine, Metahumans, Google Dialogflow, and OpenAI's Chat GPT, our system offers a novel approach to real-time, tailored interactions with virtual humans.

Future work includes the integration of real-time gestures and animations in order to enhance the immersive nature of user interactions, adding a layer of non-verbal communication to the virtual human's responses. This extension could contribute to a more natural and intuitive user experience, particularly in healthcare scenarios where non-verbal cues play a significant role.

In conclusion, our virtual human architecture represents a significant step forward in the realm of real-time, tailored interactions. By overcoming the constraints of pre-scripted interactions, our system contributes to the advancement of virtual human technology, particularly in healthcare domains such as clinical trial recruitment. We anticipate that our work will inspire further research and development in the intersection of virtual human technology, healthcare applications, and immersive user experiences.

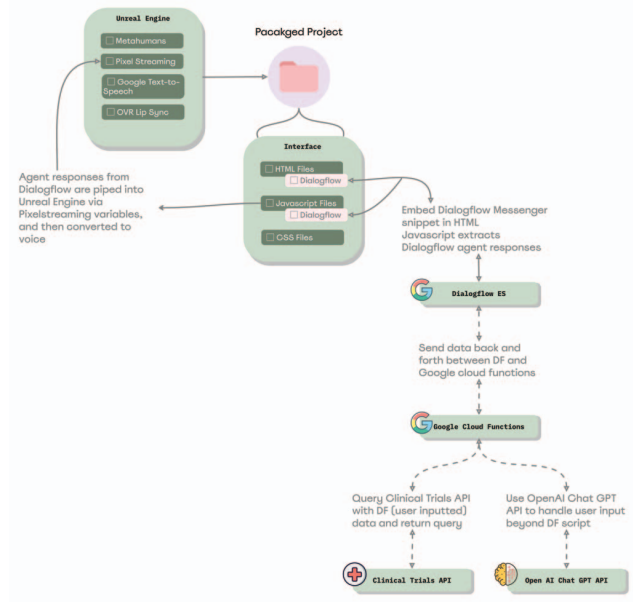


Figure 2: Overview of major technologies and how they communicate to achieve a real-time virtual human interaction

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